



Fun with IoT: Building Smart Dashboards!

Term 4 Capstone Guide & Activity Sheet for 5th Grade Explorers



Aarav's Tip: "IoT stands for **Internet of Things**! It means connecting everyday items—like water tanks, plant pots, or fan switches—to the internet so we can monitor and control them from anywhere!"

1. What Real-World Problems Are We Solving?

Imagine if your house or school could talk to you when something needs help. Here are 3 cool problems we solve with IoT:



1. Water Overflow

Water tanks on roofs often overflow, wasting electricity and thousands of liters of clean water!



2. Thirsty Crops

Farmers lose crops if soil gets too dry or too wet. Smart sensors tell us exactly when plants need water!



3. Air Pollution

Cities like New Delhi have high AQI (Air Quality Index). IoT sensors alert people when air quality gets harmful.

2. How IoT Works: The 3 Easy Layers

Just like how your body senses things, thinks, and acts, an IoT system has 3 layers:

Layer Name	What It Does (Simple Analogy)	Real Hardware Example
1. Sensor Layer (The Eyes & Ears)	Gathers information from the physical world.	Ultrasonic sensor (measures water level), Soil Moisture sensor, Air sensor.
2. Network / Cloud Layer (The Messenger)	Sends the collected data through Wi-Fi to the internet cloud.	Wi-Fi module (ESP8266/ESP32), Cloud Servers.
3. Dashboard Layer (The Brain & Screen)	Shows pictures/numbers to humans and controls devices automatically.	Interactive Dashboard screen with Gauges, Line Graphs, and Switches.



Ananya's Tip: "Each layer has a special job! Sensors collect data, Wi-Fi sends it, and Dashboards display it!"

3. Dashboard Building Blocks & Beginner Code

To view our data on a phone or laptop screen, we use three main widgets:

- **Gauge:** Like a speedometer! Great for percentages (e.g. Water Level 0% to 100%).
- **Line Graph:** Shows changes over time (e.g. Temperature or Air Quality throughout the day).
- **Switch / Toggle:** An ON/OFF button to control devices remotely (e.g. Turn Pump ON or OFF).

Beginner Code Example: Reading a Sensor & Sending Data

Here is simple Python code showing how an IoT device reads a water sensor and sends alerts:

```
# Simple 5th-Grade IoT Water Level Monitor Code
import time

def read_water_sensor():
    # Simulating water level percentage from 0% to 100%
    water_level = 85
    return water_level

# Main Loop: Check water level every few seconds
current_level = read_water_sensor()
print("Current Water Level:", current_level, "%")

if current_level >= 80:
    print("ALERT: Tank is almost FULL! 🟢")
else:
    print("Status: Water Level Normal 🟡")
```

4. Automation Logic: IF-THEN-ELSE Block Rules

We can make machines make smart decisions automatically using **IF-THEN-ELSE logic**!

```
# Example 1: Automatic Water Pump Control
IF Water_Level >= 90% THEN
    Set Pump_Status = OFF # Stop pump so water won't overflow!
ELSE IF Water_Level <= 20% THEN
    Set Pump_Status = ON # Start pump because tank is low!
ELSE
    Keep current Pump_Status
END IF
```

```
# Example 2: Plant Watering System Code
soil_moisture = 15 # Low moisture means dry soil

if soil_moisture < 30:
    print("Soil is dry! Turning Irrigation Pump ON ☁️")
else:
    print("Soil is moist. Irrigation Pump OFF ☀️")
```

5. How to Fix Problems (Debugging Checklist)

When your dashboard isn't updating, follow these 3 detective steps:

1. **Check Wires:** Are all wires connected tightly to the sensor and board?
2. **Check Wi-Fi:** Is your IoT board connected to the internet?
3. **Check Code & Rules:** Did you select the right pin or variable name in your software dashboard?



Student Worksheet & Knowledge Check

Name: _____ Date: _____ Class: 5th Grade

Question 1 (Multiple Choice)

Which layer of IoT directly senses and collects data from physical surroundings (like water or soil)?

- A) Dashboard / Application Layer
- B) Network & Cloud Layer
- C) Sensor / Perception Layer
- D) Wi-Fi Router Layer

Answer: _____

Question 2 (Short Answer)

In your own words, describe how data travels from a plant in a garden to a screen on your smartphone.

Question 3 (Widget Matching)

Match each scenario with the best dashboard widget (**Gauge**, **Line Graph**, or **Switch**):

- a) Tracking how temperature changes up and down over 24 hours: _____
- b) Turning a garden water sprinkler ON or OFF manually: _____
- c) Displaying current water fill percentage (0% to 100%) in a tank: _____

Question 4 (Block Code Logic)

Write an **IF-THEN-ELSE** rule to automatically turn a plant watering pump ON when soil moisture drops below 25%, and OFF otherwise:

IF Soil Moisture < 25% **THEN**

Set Motor = _____

ELSE

Set Motor = _____

END IF

Question 5 (Debugging Detective)

Your water level gauge on the screen stays at 0% even though the water tank is completely full! List 2 simple steps you will take to troubleshoot and fix it.

1) _____

2) _____

Question 6 (Creative Real-World Thinking)

Pick ONE problem (Water Overflow, Farm Irrigation, or Air Quality). How can a smart IoT dashboard help save resources and help your school or neighborhood?
